

BEEE – UNIT I

Short Questions

1. Define KCL.

Ans: Kirchhoff's Current Law: Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node is zero.

Mathematically, KCL implies that

$$\sum_{i=1}^N i_n = 0$$

Where N is the number of branches Connected to the node and i_n is the n^{th} current entering (or leaving) the node

2. Define KVL?

Ans: Kirchhoff's Voltage Law: Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.

Expressed mathematically, KVL states that

$$\sum_{i=1}^M v_m = 0$$

Where M is the number of voltages in the loop (or the number of branches in the loop) and V_m is the m^{th} voltage.

3. State Superposition theorem.

Ans: In any linear bilateral electrical network containing two or more independent sources, the current through or voltage across any element is the algebraic sum of the currents or voltages produced by each independent source acting alone, while all other independent sources are replaced by their internal impedances (i.e., voltage sources are short-circuited and current sources are open-circuited).

4.State Ohm's Law.

Ans: Ohm's Law states that *at constant temperature and other physical conditions, the current flowing through a conductor is directly proportional to the potential difference applied across it.*

$$V \propto I, \quad V = IR$$

Where: V = Voltage (Volts), I = Current (Amperes), R = Resistance (Ohms)

5. State any two limitations of Ohm's Law.

- Ans:**
1. Not applicable to non-linear or unilateral devices.
 2. Not Valid for High Voltages.
 3. Not Applicable to Non-Metallic Conductors

6. Define the term Rms value.

Ans: RMS value is defined as the square root of the area under the squared curve divided by the base line of the curve.

$$\text{RMS value} = \sqrt{\frac{\text{Area under squared curve}}{\text{Base line}}}$$

(or)

The Root Mean Square (RMS) value of an alternating current is the steady value of direct current (DC) which when flowing through a given circuit for a given time, produces the same amount of heat as would be produced by the alternating current flowing in the circuit for the same time.

7. Define Average value.

Ans: The average of all instantaneous values of an alternating quantity over one complete cycle is called average value.

(or)

Average value is defined as area under of the curve divided by the time period of the curve.

$$\text{Average value} = \frac{\text{Area under the curve}}{\text{Time Period}}$$

8. Define Form Factor.

Ans: Form factor is defined as the ratio of RMS value to average value of an alternating quantity.

$$\text{Form factor} = \frac{\text{RMS value}}{\text{Average value}}$$

9. Define Peak Factor.

Ans: Peak factor is defined as the ratio of Maximum value to RMS value of an alternating quantity.

$$\text{Peak factor} = \frac{\text{Maximum value}}{\text{RMS value}}$$

10. Define Impedance.

Ans: Impedance is the total opposition offered by an AC circuit to the flow of alternating current.

It includes the effect of resistance and reactance.

Expression

$$Z = R + jX$$

For magnitude:

$$|Z| = \sqrt{R^2 + X^2}$$

Where:

R = resistance (Ω), X = reactance (Ω)

Inductive reactance: $X_L = \omega L$, Capacitive reactance: $X_C = \frac{1}{\omega C}$

11. Define Power Factor.

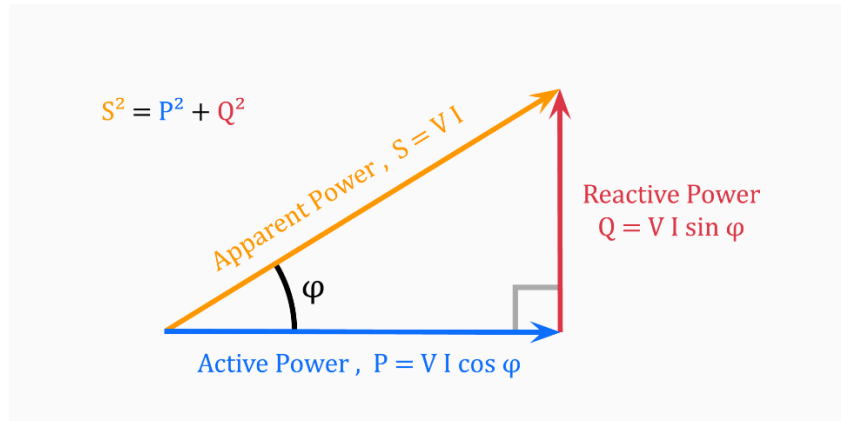
Ans: Power factor is the cosine of the phase angle between voltage and current.

$$\text{Power factor} = \cos \phi = \frac{R}{Z} \quad (\text{or}) \quad \text{Power factor} = \cos \phi = \frac{P}{S}$$

12. Draw the power triangle.

Ans: Power Triangle (Relationship):

- Base → Active power (P)
- Vertical → Reactive power (Q)
- Hypotenuse → Apparent power (S)



$$S^2 = P^2 + Q^2$$

13. Define Amplitude, time period, and frequency of an AC wave form.

Ans:

Amplitude (Peak Value): Amplitude is the maximum value attained by an AC voltage or current from the zero reference.

- For voltage → V_m
- For current → I_m

Time Period (T): Time period is the time taken by an AC waveform to complete one full cycle.

$$T = \frac{1}{f}$$

Where:

- T = time period (seconds)
- f = frequency (Hz)

Frequency (f): Frequency is the number of cycles completed by the AC waveform in one second. Unit of frequency is **Hertz (Hz)**

$$f = \frac{1}{T}$$

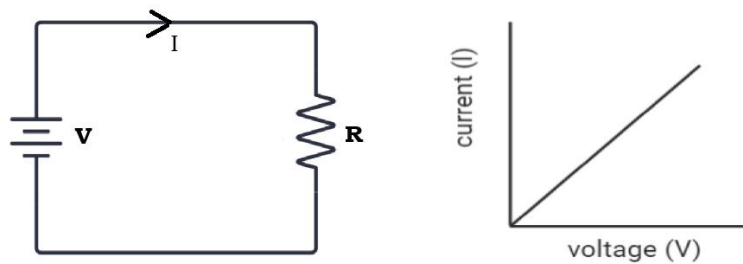
In India, standard supply frequency = **50 Hz**

Long Questions

1. What is ohm's law, give its limitations and applications?

Ans: Ohm's law states that at constant temperature and physical conditions, the current (I) flowing through a conductor is directly proportional to the potential difference (V) applied across it.

$$V \propto I$$



Introducing a proportionality constant called **resistance (R)**,

$$V = IR$$

Where

- V = potential difference (volts)
- I = current (amperes)
- R = resistance (ohms, Ω)

V-I Characteristics:

- ✓ A graph is plotted between voltage (V) on the X-axis and current (I) on the Y-axis.
- ✓ For an ohmic conductor, the graph is a straight line passing through the origin.
- ✓ The slope of the V-I graph represents the resistance of the conductor.

Limitations of Ohm's Law:

1. Ohm's Law applies only to ohmic materials (like copper, aluminium) where resistance remains constant.
2. It does not apply to non-linear devices like Diodes, Transistors, Thermistors and Electric arc lamps
3. Ohm's Law is valid only if physical conditions remain constant.
4. Not Applicable to Unilateral Devices.

5. Not Valid for High Voltages.
6. Not Applicable to Non-Metallic Conductors.

Ohm's Law is valid only for linear, metallic conductors under constant temperature and physical conditions. It is not applicable to non-linear, unilateral, or temperature-dependent devices.

2. Explain Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Current (KVL) Law by considering one example.

Ans:

Kirchhoff's Current Law: Kirchhoff's current law (KCL) states that the algebraic sum of currents (entering and leaving) at a node is zero.

Mathematically, KCL implies that

$$\sum_{i=1}^N i_n = 0$$

Where N is the number of branches Connected to the node and i_n is the n^{th} current entering (or leaving) the node.

Consider five conductors, carrying currents I_1, I_2, I_3, I_4 and I_5 meeting at a point O as shown in Fig. Assuming the entering currents to be positive and leaving currents negative, we have

Applying KCL at point O, gives

$$i_1 + (-i_2) + i_3 + i_4 + (-i_5) = 0$$

Since currents i_1, i_3 , and i_4 are entering the node, while currents i_2 and i_5 are leaving it.

By rearranging the terms, we get

$$i_1 + i_3 + i_4 = i_2 + i_5$$

The sum of the currents entering a node is equal to the sum of the currents leaving the node.

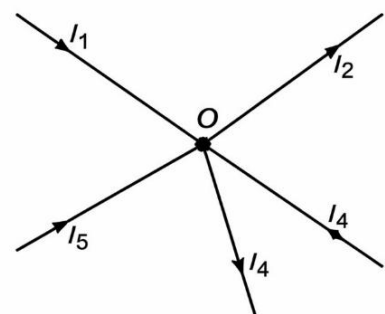


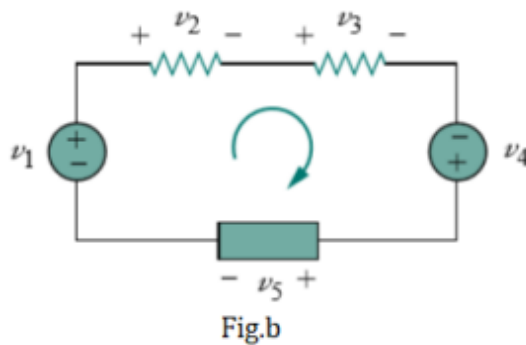
Fig: Kirchhoff's Current Law

Kirchhoff's Voltage Law: Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.

Expressed mathematically, KVL states that

$$\sum_{i=1}^M v_m = 0$$

Where M is the number of voltages in the loop (or the number of branches in the loop) and V_m is the m^{th} voltage.



We can start with any branch and go around the loop either clockwise or counterclockwise. Suppose we start with the voltage source and go clockwise around the loop as shown, then voltages would be $+V_1$, $-V_2$, $-V_3$, $+V_4$, and $-V_5$, in that order.

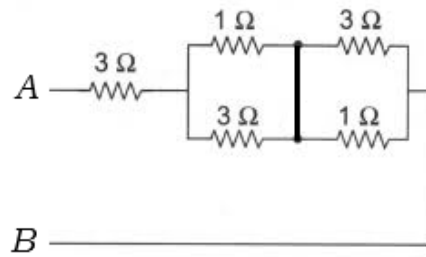
$$V_1 - V_2 - V_3 + V_4 - V_5 = 0$$

Rearranging terms gives, $V_1 + V_4 = V_2 + V_3 + V_5$

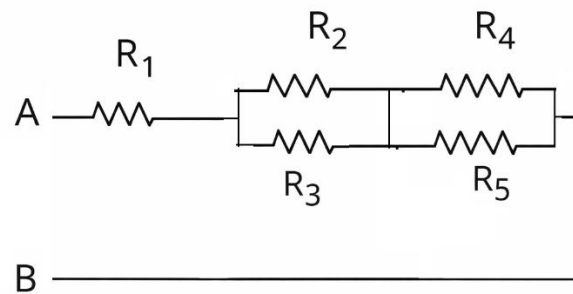
Which may be interpreted as

The sum of the voltage drops around a loop is equal to the sum of the voltage rises around the loop.

3. Find the equivalent resistance for the given circuit.



Ans:



In the above circuit R_2 and R_3 , R_4 and R_5 are in parallel.

Calculation of R_X :

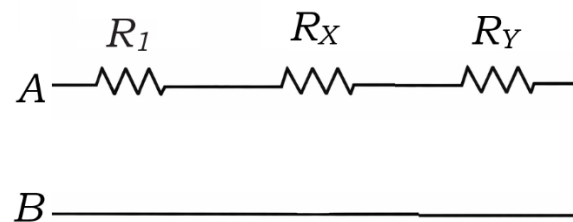
$$R_X = \frac{R_2 R_3}{R_2 + R_3} = \frac{1 \times 3}{1 + 3}$$

$$R_X = \frac{3}{4} \Omega$$

Calculation of R_Y :

$$R_Y = \frac{R_4 R_5}{R_4 + R_5} = \frac{3 \times 1}{3 + 1}$$

$$R_Y = \frac{3}{4} \Omega$$

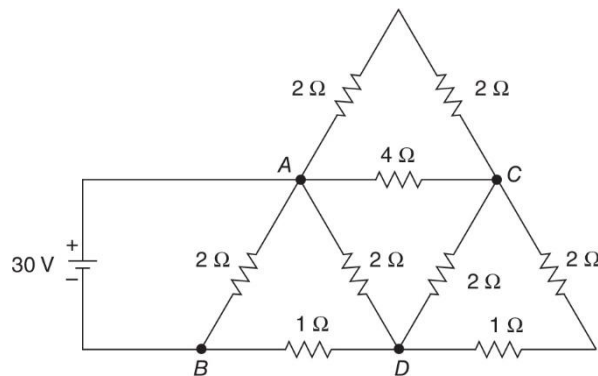


In the above circuit R_1 , R_X and R_Y are series

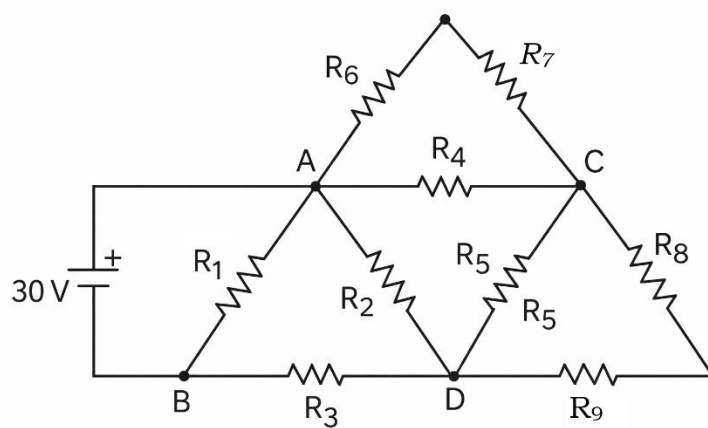
$$R_{AB} = R_1 + R_X + R_Y = 3 + (3/4) + (3/4) = 4.5\Omega$$

4. Determine the current delivered by the source in the circuit shown.

Ans:



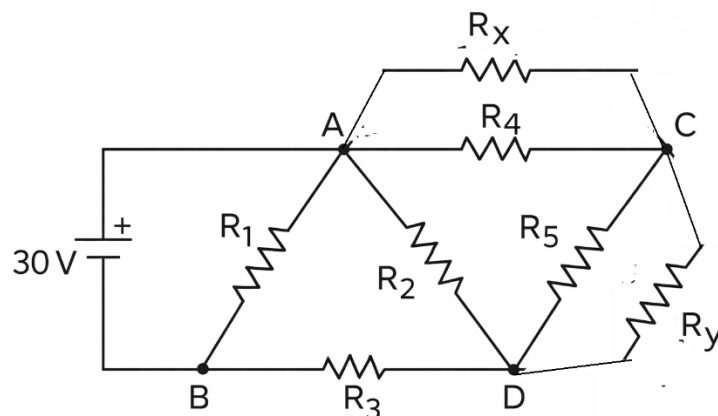
Ans:



In the above circuit R_6 and R_7 , R_8 and R_9 are in series,

$$R_X = R_6 + R_7 = 2 + 2 = 4\Omega,$$

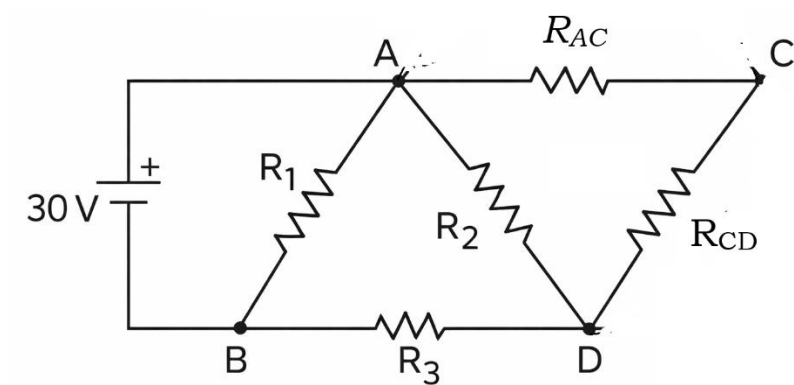
$$R_Y = R_8 + R_9 = 2 + 1 = 3\Omega$$



In the above circuit R_4 and R_X , R_5 and R_Y are in parallel

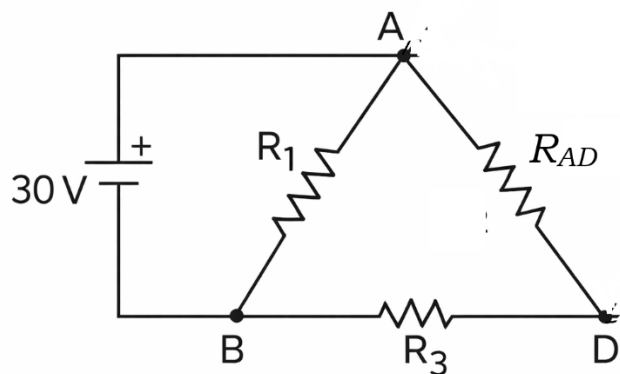
$$R_{AC} = \frac{R_X R_4}{R_X + R_4} = \frac{4 \times 4}{4 + 4} = 2\Omega$$

$$R_{CD} = \frac{R_Y R_5}{R_Y + R_5} = \frac{3 \times 2}{3 + 2} = 1.2 \Omega$$



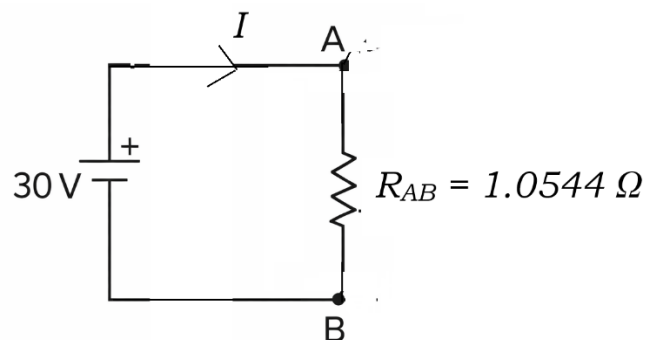
In the above circuit R_{AC} and R_{CD} are series, resultant is in parallel with R_2

$$R_{AD} = \frac{(R_{AC} + R_{CD}) R_2}{(R_{AC} + R_{CD}) + R_2} = \frac{(2 + 1.2) 2}{(2 + 1.2) + 2} = \frac{6.4}{5.2} = 1.231 \Omega$$



In the above circuit R_{AD} and R_3 are in series, resultant is in parallel with R_1

$$R_{AB} = \frac{(R_{AD} + R_3) R_1}{(R_{AD} + R_3) + R_1} = \frac{(1.23 + 1) 2}{(1.23 + 1) + 2} = \frac{4.46}{4.23} = 1.0544 \Omega$$



$$\text{Current } I = \frac{V}{R_{AB}} = \frac{30}{1.0544} = 28.45 \text{ A}$$

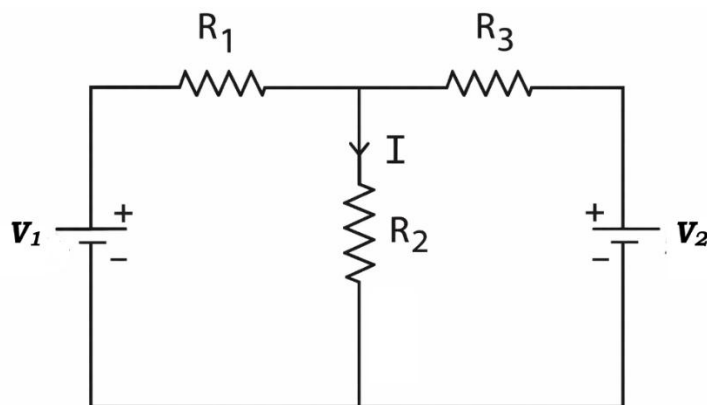
5. State Superposition theorem and explain about it with suitable example.

Ans: In any linear bilateral electrical network containing two or more independent sources, the current through or voltage across any element is the algebraic sum of the currents or voltages produced by each independent source acting alone, while all other independent sources are replaced by their internal impedances (i.e., voltage sources are short-circuited and current sources are open-circuited).

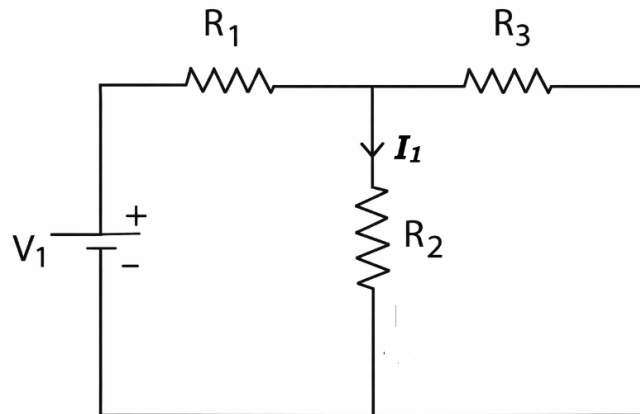
Superposition theorem is applicable only to linear networks (both AC and DC) where current is linearly related to voltage as per Ohm's law.

Example:

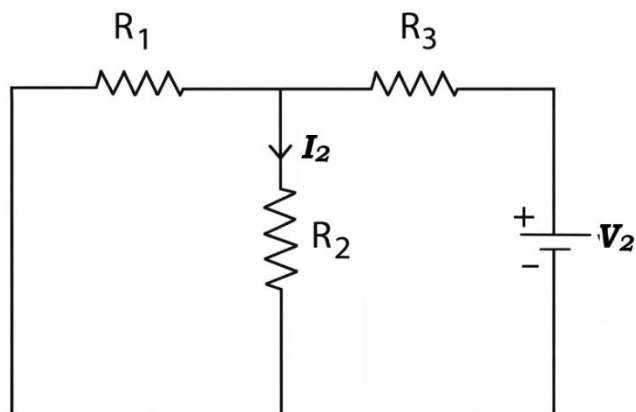
Find the current (I) for the given circuit using superposition theorem.



Step 1: Find the current (I_1) when voltage source (V_1) acting alone and other sources are replaced with their internal resistance.



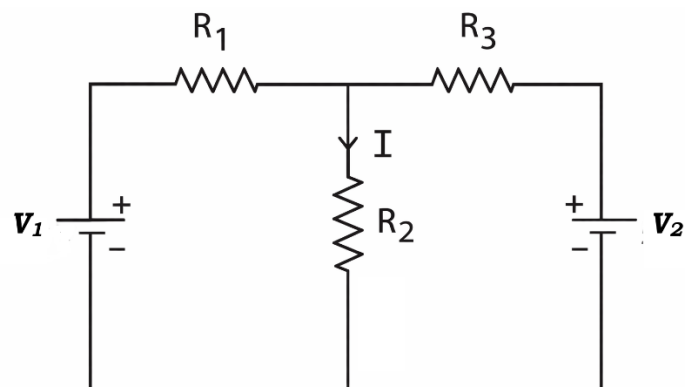
Step 2: Find the current (I_2) when voltage source (V_2) acting alone and other sources are replaced with their internal resistance.



As per superposition theorem current in the load resistance is $= I_1 + I_2$

Step 3: verification

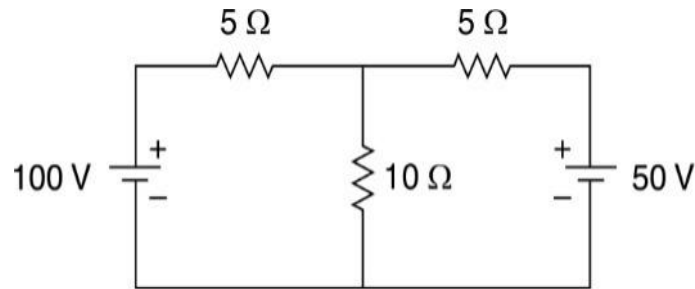
Find the current (I) when voltage sources V_1 and V_2 acting.



Hence current through the R_2 resistor is when both the sources are acting simultaneously,

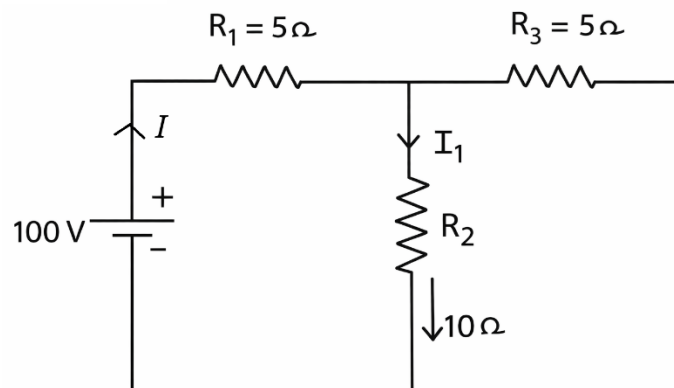
As per superposition theorem , $I = I_1 + I_2$

6. Compute the current in the 10Ω resistor as shown in Fig. using Superposition theorem.



Ans:

Step 1: Find the current (I_1) when voltage source $V_1 = 100V$ acting alone and other sources are replaced with their internal resistance.



$$R_{eq} = R_1 + \frac{R_2 R_3}{R_2 + R_3}$$

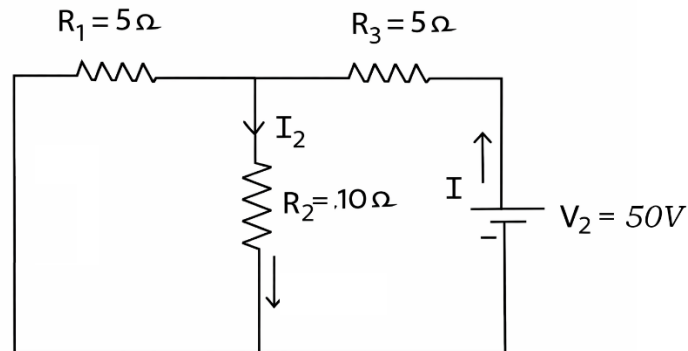
$$R_{eq} = 5 + \frac{(10)(5)}{10 + 5} = 5 + \frac{50}{15} = 5 + 3.33$$

$$R_{eq} = 8.33 \Omega$$

$$I = \frac{V}{R_{eq}} = \frac{100}{8.33} = 12A$$

Hence current through 10 Ω resistor, $I_1 = I \frac{R_3}{R_2 + R_3} = 12 \frac{5}{10 + 5} = 4\text{A}$

Step 2: Find the current (I_2) when voltage source **$V_2 = 50\text{V}$** acting alone and other sources are replaced with their internal resistance.



$$R_{eq} = R_3 + \frac{R_1 R_2}{R_1 + R_2}$$

$$R_{eq} = 5 + \frac{(5)(10)}{5 + 10} = 5 + \frac{50}{15} = 5 + 3.33$$

$$R_{eq} = 8.33 \Omega$$

$$I = \frac{V}{R_{eq}} = \frac{50}{8.33} = 6\text{A}$$

Hence current through 10 Ω resistor, $I_2 = I \frac{R_1}{R_1 + R_2} = 6 \frac{5}{5 + 10} = 2\text{A}$

As per superposition theorem current in the load resistance is = **$I_1 + I_2$**

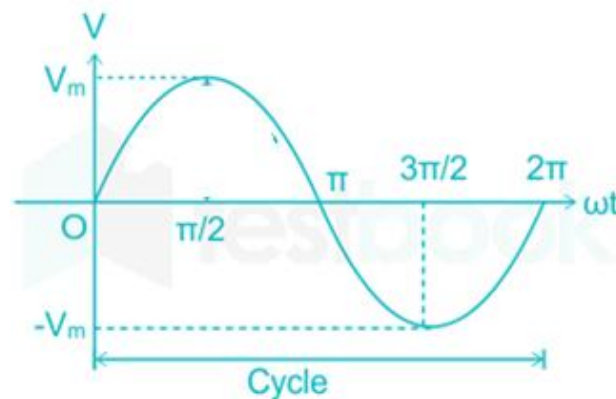
$$= 4 + 2 = 6\text{A}$$

7. Derive the average value, R.M.S. value, form factor and Peak factor of a sinusoidal alternating voltage.

Ans:

Average value for sinusoidal voltage wave-form: A sine wave is symmetrical, its area below horizontal axis is the same as its area above the axis, thus, over a full cycle its net area is zero.

The average of half a sine wave, however, is not zero. So calculate average value for sinusoidal wave form only for half cycle.



$$V_{\text{avg}} = \frac{\text{Area under the half cycle}}{\text{Time period of half cycle}}$$

$$\begin{aligned} V_{\text{avg}} &= \frac{1}{\pi} \int_0^{\pi} v \, d(\omega t) \\ &= \frac{1}{\pi} \int_0^{\pi} V_m \sin(\omega t) \, d(\omega t) \\ &= \frac{V_m}{\pi} [-\cos(\omega t)]_0^{\pi} \\ &= \frac{V_m}{\pi} [(-\cos \pi) - (-\cos 0)] \\ &= \frac{V_m}{\pi} [1 + 1] \\ &= \frac{2V_m}{\pi} \\ \boxed{V_{\text{avg}} &= \frac{2V_m}{\pi} = 0.637 V_m} \end{aligned}$$

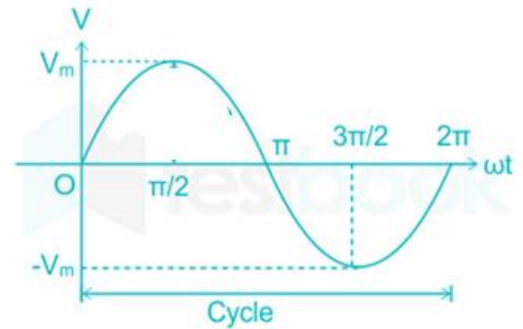
RMS value: RMS value is defined as the square root of the area under the squared curve divided by the base line of the curve.

$$\text{RMS value} = \sqrt{\frac{\text{Area under squared curve}}{\text{Base line}}}$$

Derivation of RMS Value of Sinusoidal Voltage:

$$\begin{aligned} V_{\text{RMS}} &= \sqrt{\frac{1}{T} \int_0^T v^2 dt} \\ &= \sqrt{\frac{1}{2\pi} \int_0^{2\pi} (V_m \sin \omega t)^2 d(\omega t)} \\ &= \sqrt{\frac{1}{2\pi} \int_0^{2\pi} V_m^2 \sin^2 \omega t d(\omega t)} \\ &= \sqrt{\frac{V_m^2}{2\pi} \int_0^{2\pi} \frac{1 - \cos 2\omega t}{2} d(\omega t)} \\ &= \sqrt{\frac{V_m^2}{4\pi} \left[\omega t - \frac{\sin 2\omega t}{2} \right]_0^{2\pi}} \\ &= \sqrt{\frac{V_m^2}{4\pi} [(2\pi - 0)]} \\ &= \sqrt{\frac{V_m^2}{2}} \end{aligned}$$

$$\boxed{V_{\text{RMS}} = \frac{V_m}{\sqrt{2}}}, \quad \mathbf{V_{RMS} = 0.707V_m.}$$



Form Factor: Form factor is defined as the ratio of RMS value to average value of an alternating quantity.

Formula

$$\text{Form factor} = \frac{\text{RMS value}}{\text{Average value}}$$

For a sinusoidal waveform

$$\text{Form factor} = \frac{V_{\text{RMS}}}{V_{\text{avg}}} = \frac{0.707V_m}{0.637V_m} = 1.11$$

8. An alternating current is represented by $i = 50 \sin 314 t$. Determine

(i) Average value (iii) Peak factor (ii) RMS value (iv) Form factor.

Ans:

Given alternating current: $i = 50 \sin (314t)$

This is in standard form

$$i = I_m \sin (\omega t)$$

So,

$$I_m = 50 \text{ A (maximum or peak current)}$$

(i) Average Value I_{avg}

For a sinusoidal AC, average value is taken over half cycle:

$$\begin{aligned} I_{avg} &= \frac{2I_m}{\pi} \\ I_{avg} &= \frac{2 \times 50}{\pi} \\ I_{avg} &= \frac{100}{3.142} \approx 31.83 \text{ A} \end{aligned}$$

(ii) RMS Value I_{rms}

$$\begin{aligned} I_{rms} &= \frac{I_m}{\sqrt{2}} \\ I_{rms} &= \frac{50}{\sqrt{2}} = \frac{50}{1.414} \\ I_{rms} &\approx 35.36 \text{ A} \end{aligned}$$

(iii) Peak Factor (Crest Factor)

$$\begin{aligned} \text{Peak Factor} &= \frac{I_m}{I_{rms}} \\ &= \frac{50}{35.36} \\ \text{Peak Factor} &\approx 1.414 \end{aligned}$$

(iv) Form Factor

$$\begin{aligned} \text{Form Factor} &= \frac{I_{rms}}{I_{avg}} \\ &= \frac{35.36}{31.83} \\ \text{Form Factor} &\approx 1.11 \end{aligned}$$

9 .What are the passive elements in the network? Discuss them in detail with examples.

Ans: A passive element is an electrical component that cannot generate electrical energy. It can only absorb, dissipate, or store energy supplied from an external source.

There are three basic passive elements in electrical engineering:

- 1.Resistor (R) 2.Inductor (L) 3.Capacitor (C)

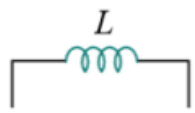
Resistor: A resistor is a passive element that is used to limit (or) regulate the flow of electric current.

Resistance: The resistance R of an element denotes its ability to resist the flow of electric current. It is measured in ohms (Ω).

$$R = \frac{v}{i}$$

Power observed by the resistor is $p = vi = i^2 R = \frac{v^2}{R}$

Inductor: An inductor is a passive element designed to store energy in its magnetic field. An inductor consists of a coil of conducting wire. If current is made to pass through an inductor, an electromagnetic field is formed. A change in current, produces change in the electromagnetic field, which induces a voltage across the coil according to faraday's law of electromagnetic induction.



A circuit diagram showing an inductor represented by a green coil symbol. Above the coil is the letter 'L'. The coil is connected to two horizontal lines representing wires.

$$v = L \frac{di}{dt}$$

Where

V = voltage across inductor in volts,

i = current flowing through inductor in amps

Inductance is the property of a material, by which it opposes any sudden change of current passing through it, measured in henry (H).

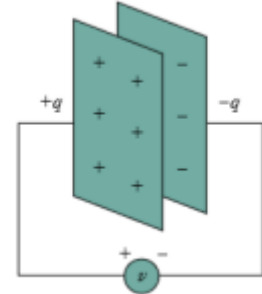
Power observed by inductor is

$$p = vi = L \left(\frac{di}{dt} \right) i$$

Energy stored by the inductor is $w = \frac{1}{2} L i^2$

Capacitor: A capacitor is a passive element designed to store energy in its electric field. Capacitors are used extensively in electronics, communications, computers, and power systems.

A capacitor consists of two conducting plates separated by an insulator (or dielectric). When a voltage source v is connected to the capacitor, as in Fig., the source deposits a positive charge $+q$ on one plate and a negative charge $-q$ on the other. The capacitor is said to store the electric charge.



The amount of charge stored, represented by q , is directly proportional to the applied voltage v so that

$$q = C * v$$

$$\Rightarrow C = \frac{q}{v}$$

Capacitance is a measure of the amount of electric charge stored for a given electric potential and is given by the ratio of the charge on one plate of a capacitor to the voltage difference between the two plates, measured in farads (F).

Power observed by capacitor is

$$p = vi = v * C \left(\frac{dv}{dt} \right) \left(\because i = C \frac{dv}{dt} \right)$$

Energy stored by the capacitor is

$$w = \frac{1}{2} C v^2$$

10. Show the voltage and current relationship with phasor diagrams in R, L and C circuits?

Ans: Resistor: A simple circuit containing a resistor R and energised by an alternating voltage varying sinusoidally as given by $V(t) = V_m \sin(\omega t)$.

The relation for the current at any instant can be written as,

$$i = \frac{V}{R} = \frac{V_m \sin \omega t}{R} = I_m \sin \omega t$$

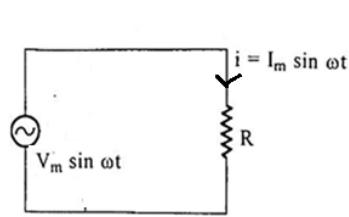


Fig. Pure resistive circuit

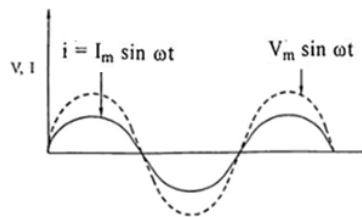


Fig. V-I waveforms for R-load

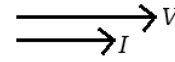


Fig. Phasor representation

From the voltage and current equations, it is observed that in a pure resistor the voltage and current are in phase.

Inductor: A simple circuit containing an inductor L and energised by an alternating voltage $V(t) = V_m \sin(\omega t)$.

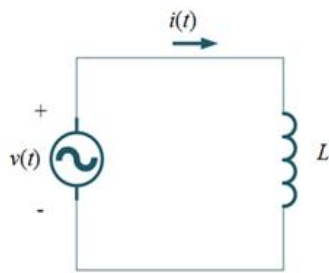


Figure : Pure inductive circuit

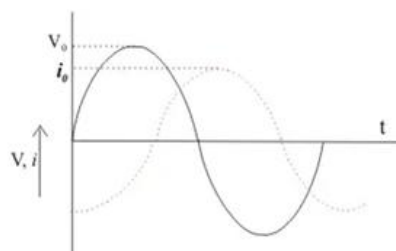


Figure : Sinusoidal representation

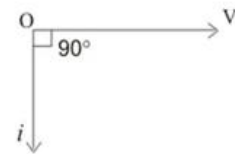


Figure : Phasor diagram representation

For an ideal inductor (no resistance),

$$v(t) = L \frac{di(t)}{dt}$$

Assume sinusoidal voltage:

$$v(t) = V_m \sin(\omega t)$$

$$i(t) = \int \frac{v(t)}{L} dt = \frac{V_m}{\omega L} (-\cos \omega t)$$

$$i(t) = I_m \sin(\omega t - 90^\circ)$$

So,

$$\boxed{i(t) = I_m \sin(\omega t - 90^\circ)}$$

Current is delayed by 90° .

From the voltage and current equations, it is observed that in a pure inductor the current lags the voltage by an angle of 90° .

Capacitor: A simple circuit containing a Capacitor “C” and energised by an alternating voltage $V(t) = V_m \sin(\omega t)$.

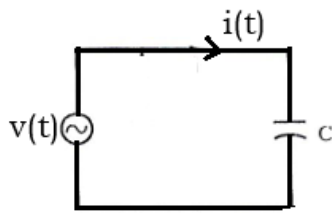


Fig. Pure capacitive circuit

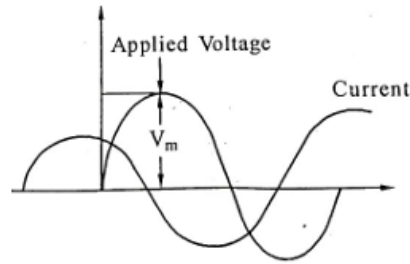


Fig. V-I waveforms for L-load

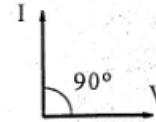


Fig. Phasor diagram

Let the applied voltage be:

$$v(t) = V_m \sin(\omega t)$$

Now differentiate:

$$i(t) = C \frac{d}{dt} [V_m \sin(\omega t)]$$

$$i(t) = CV_m \omega \cos(\omega t)$$

Using identity:

$$\cos(\omega t) = \sin(\omega t + 90^\circ)$$

$$i(t) = \omega CV_m \sin(\omega t + 90^\circ)$$

Let

$$I_m = \omega CV_m$$

$$i(t) = I_m \sin(\omega t + 90^\circ)$$

From the voltage and current equations, it is observed that in a pure capacitor the current leads the voltage by an angle of 90° .

11. Explain the terms Active, Reactive, Apparent powers and Power factor.

Ans:

1. Active Power (P): Active power is the actual power consumed in the circuit to do useful work (like heating, lighting, mechanical output). It is also called as true power and real power. Unit for active power is Watts or KW

$$P = VI \cos \phi$$

$$\text{Also written as: } P = I^2 R$$

2. Reactive Power (Q): Reactive power is the power that alternately stores and returns energy in inductors and capacitors. It does no useful work. Unit for reactive power is Volt-Ampere Reactive (VAR) or KVAR.

$$\text{Expression: } Q = VI \sin \phi$$

3. Apparent Power (S): Apparent power is the product of RMS voltage and RMS current in an AC circuit. Unit for apparent power is Volt-Ampere (VA) or KVA

$$S = VI$$

Also:

$$S = \sqrt{P^2 + Q^2}$$

4. Power Factor (cos ϕ): Power factor is the cosine of the phase angle between voltage and current.

$$\text{Power factor} = \cos \phi = \frac{R}{Z}$$

(or)

$$\text{Power factor} = \cos \phi = \frac{P}{S}$$

12. A electric circuit contains 200V, 6A, 0.5p.f supply, then find (i) Active power, (ii) Reactive power, (iii) Apparent power

Ans:

Given:

- Voltage $V = 200 \text{ V}$
- Current $I = 6 \text{ A}$
- Power factor $\cos \phi = 0.5$

(i) Active Power (Real Power) P

$$P = VI \cos \phi$$

$$P = 200 \times 6 \times 0.5$$

$$P = 600 \text{ W}$$

(ii) Reactive Power Q

$$Q = VI \sin \phi$$

First find $\sin \phi$:

$$\cos \phi = 0.5$$

$$\sin \phi = \sqrt{1 - \cos^2 \phi} = \sqrt{1 - 0.25} = \sqrt{0.75} = 0.866$$

$$Q = 200 \times 6 \times 0.866$$

$$Q = 1039.2 \text{ VAR} \approx 1.04 \text{ kVAR}$$

(iii) Apparent Power S

$$S = VI$$

$$S = 200 \times 6$$

$$S = 1200 \text{ VA} = 1.2 \text{ kVA}$$

13. Two impedances $(100 + j60)$ and $(4 - j5)$ are connected in series. Determine (i). The equivalent impedance (ii). Power factor.

Ans:

Given impedances in series:

$$Z_1 = (100 + j60) \Omega$$

$$Z_2 = (4 - j5) \Omega$$

(i) Equivalent Impedance Z_{eq}

In series connection, impedances add directly:

$$Z_{eq} = Z_1 + Z_2$$

$$Z_{eq} = (100 + j60) + (4 - j5)$$

$$Z_{eq} = (100 + 4) + j(60 - 5)$$

$$Z_{eq} = 104 + j55 \Omega$$

$$|Z| = \sqrt{104^2 + 55^2}$$

$$|Z| = \sqrt{10816 + 3025} = \sqrt{13841}$$

$$|Z| \approx 117.7 \Omega$$

(ii) Power Factor:

$$\text{Power factor} = \cos \Phi = R/Z = (104/117.7) = 0.883 \text{ lagging}$$